Hack-o-week Projects

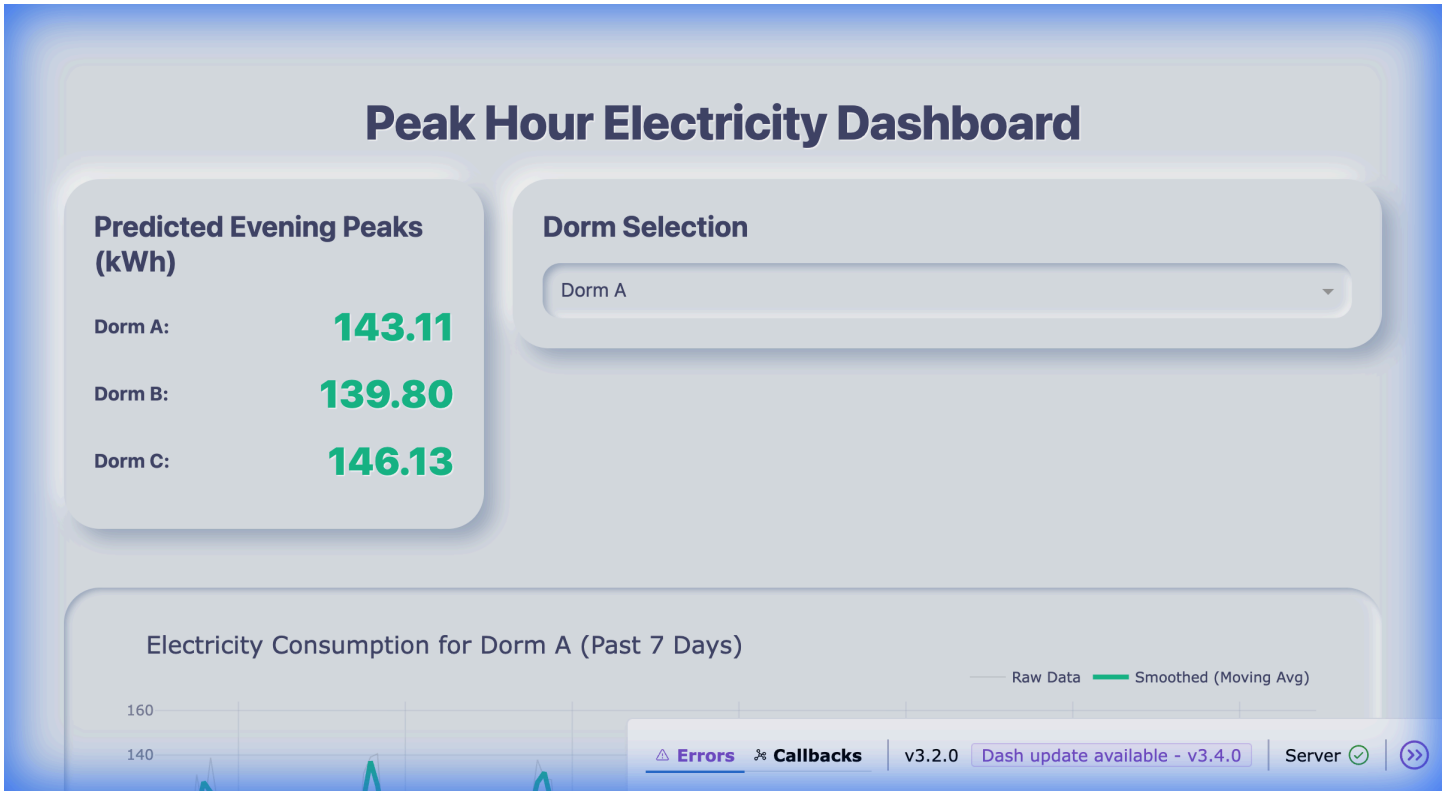
Consolidated Energy & Utility Forecasting Solutions

Date: 2026-03-12

Week 1: Peak Hour Electricity Spikes

Collect hourly meter data from dorms; apply moving average smoothing and linear regression to predict evening peaks based on past week. Visualize trends on a live Plotly dashboard.

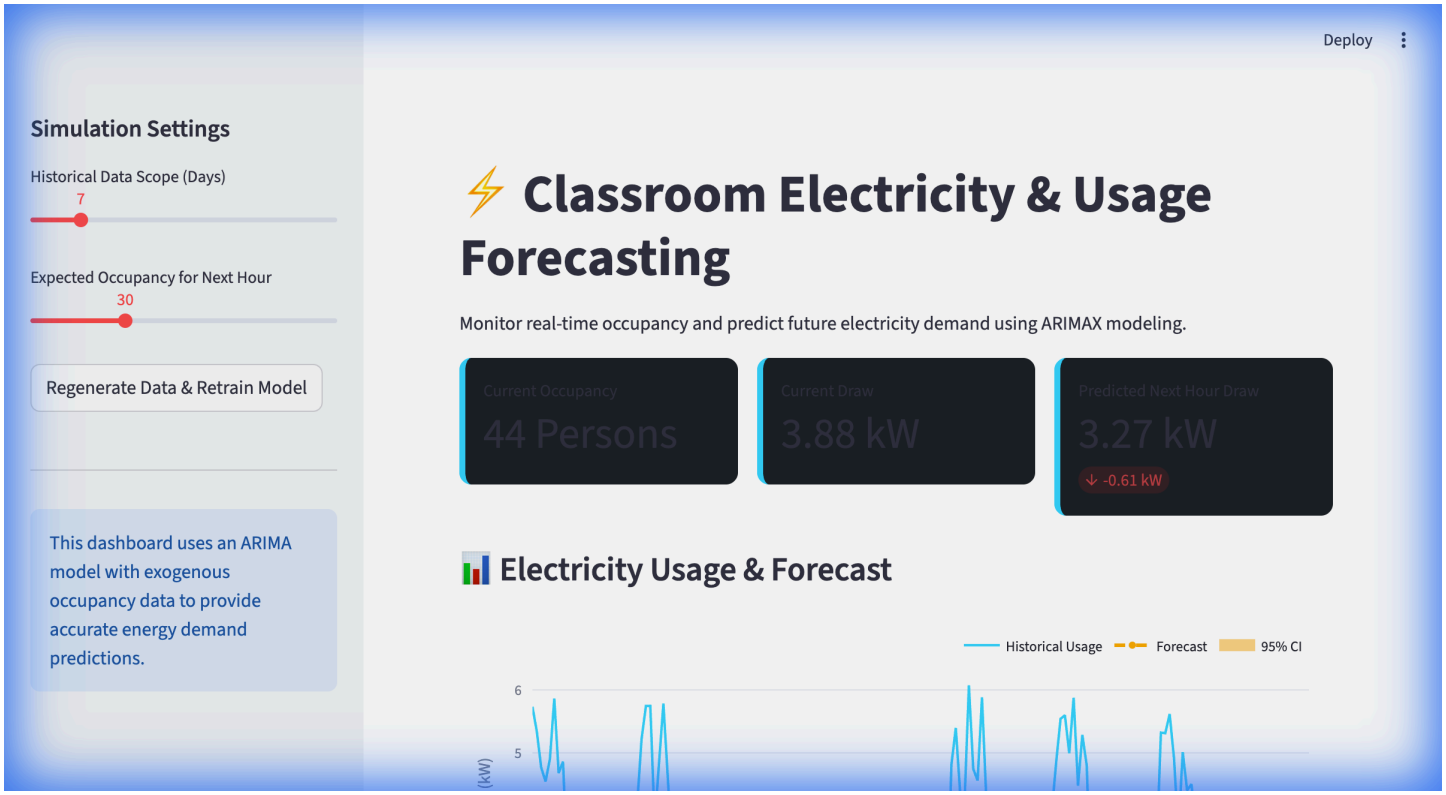
Repository: https://github.com/DevDhapodkar/Peak-Hour-Electricity-Spikes-Prediction-Dashboard



Week 2: Classroom Usage Forecasting

Use sensor data (occupancy via Wi-Fi logs) to train a simple ARIMA model for next-hour room electricity draw; dashboard shows confidence intervals.

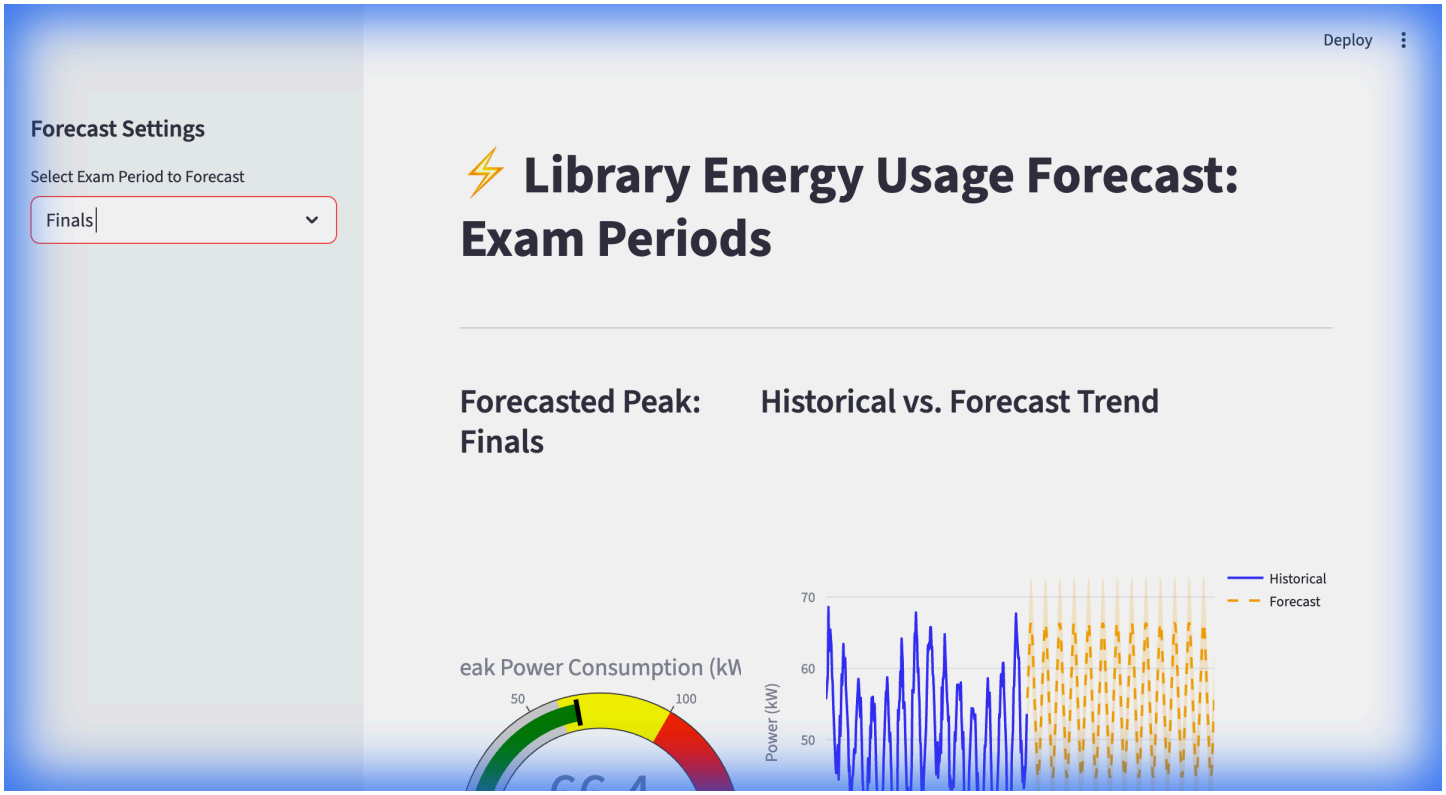
Repository: https://github.com/DevDhapodkar/Classroom-Usage-Forecasting-Walkthrough



Week 3: Library Energy During Exams

Aggregate historical usage with event calendars; implement exponential smoothing for semester-end forecasts, displayed as a gauge on Streamlit dashboard.

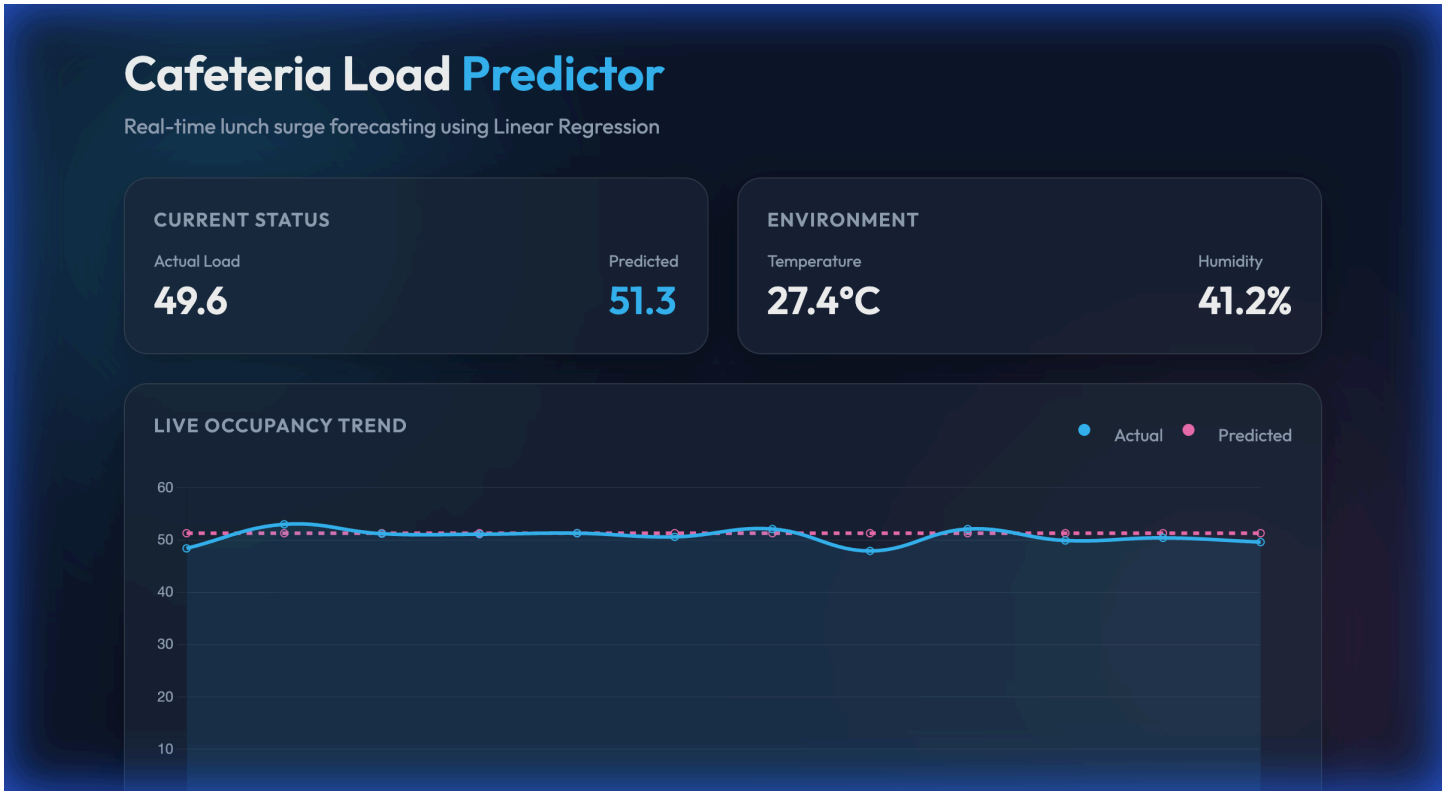
Repository: https://github.com/DevDhapodkar/Library-Energy-Forecast-During-Exams



Week 4: Cafeteria Load Prediction

Predict lunch-hour surges using temperature/weather data and linear regression; real-time line chart updates via WebSocket integration.

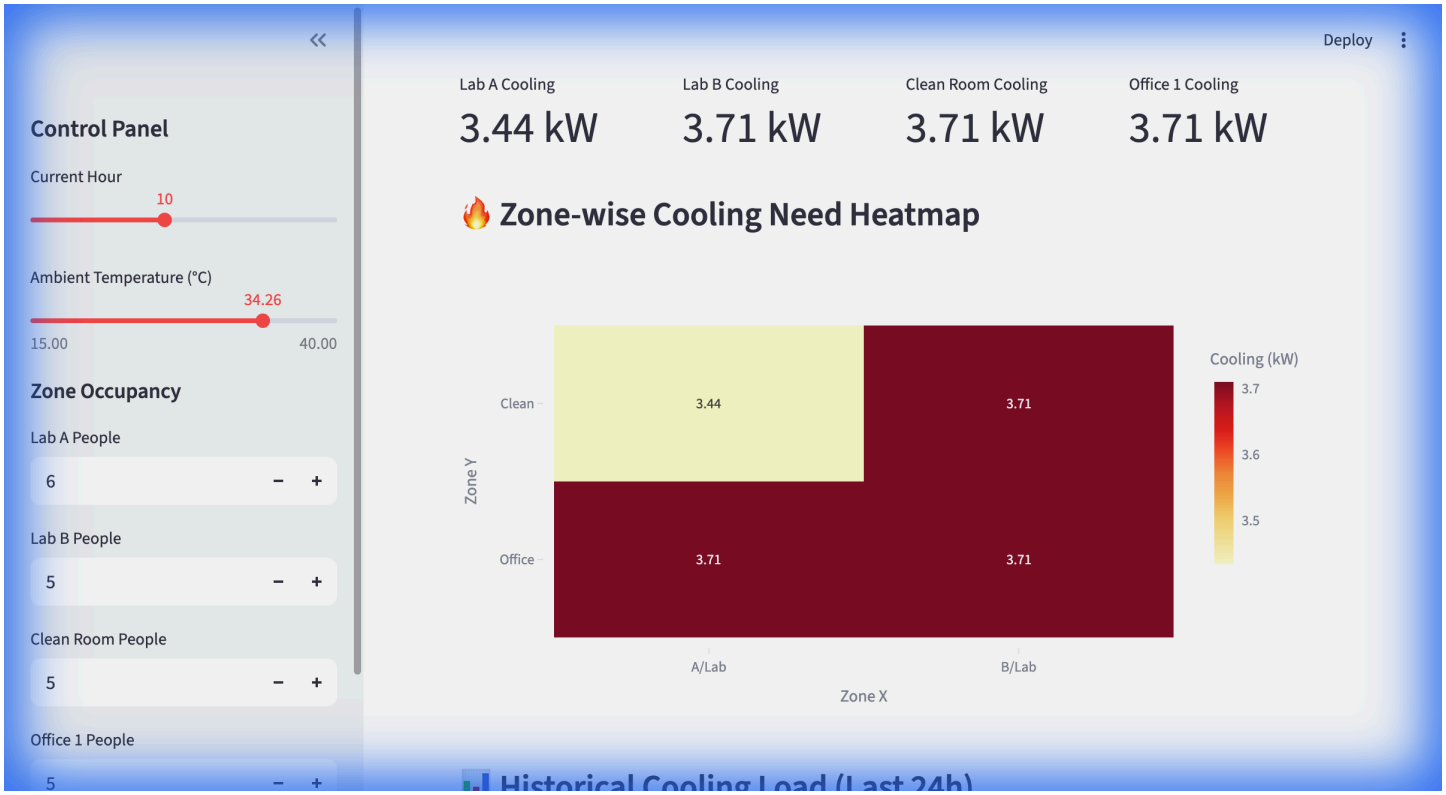
Repository: https://github.com/DevDhapodkar/Cafeteria-Load-Prediction-Tasks



Week 5: HVAC Optimization in Labs

Train a basic decision tree on occupancy/temperature data to forecast cooling needs; dashboard with heatmaps for zone-wise predictions.

Repository: https://github.com/DevDhapodkar/HVAC-Optimization-in-Labs



Week 6: Sports Facility Night Usage

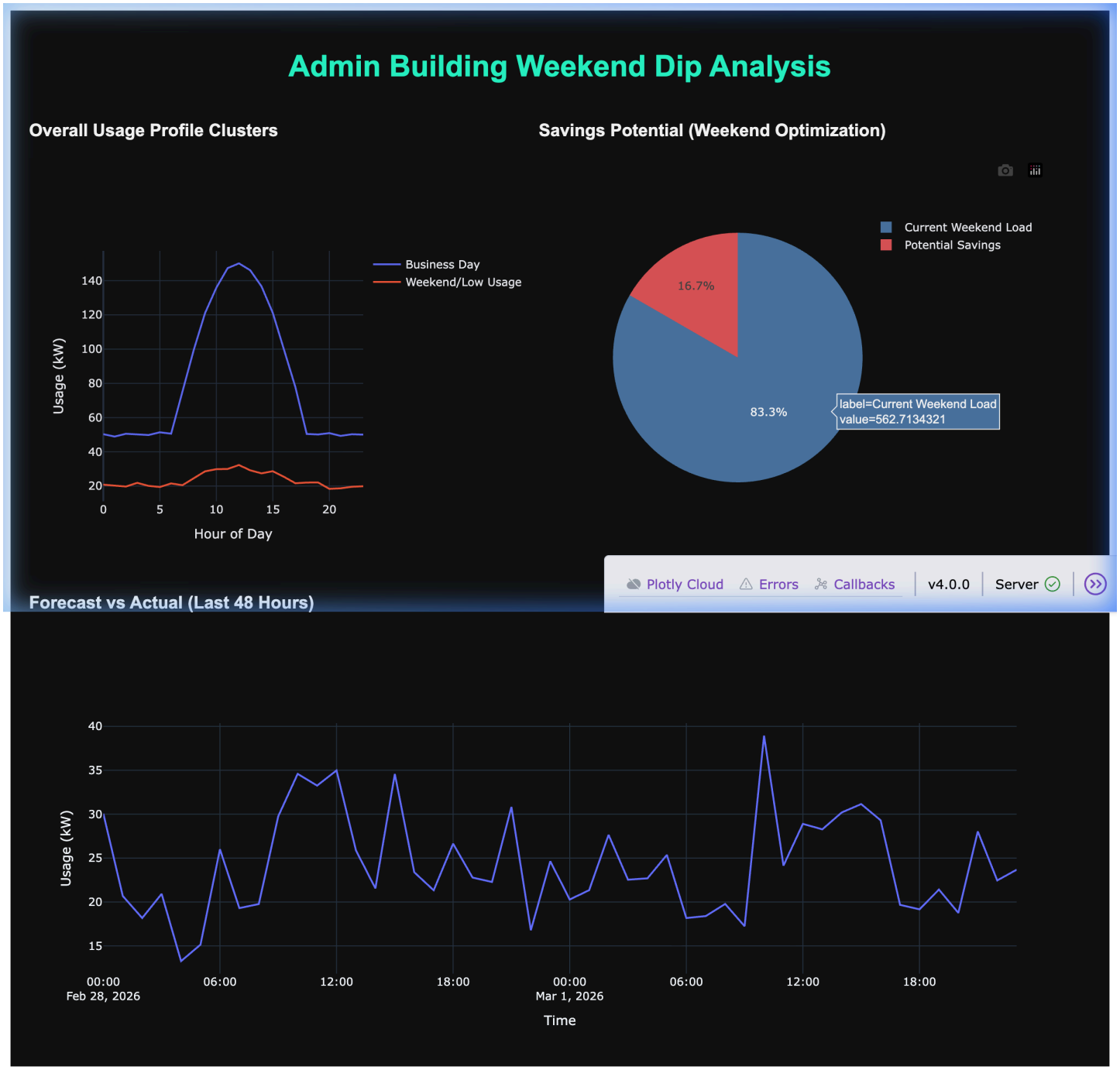
Use RNN (simple LSTM) on hourly patterns to predict post-event electricity; interactive dashboard filters by day type.

Repository: https://github.com/DevDhapodkar/Sports-Facility-Night-Usage

Week 7: Admin Building Weekend Dip

Apply k-means clustering on usage profiles then regress clusters for forecasts; pie charts show savings potential on dashboard.

Repository: https://github.com/DevDhapodkar/Admin-Building-Weekend-Dip



Week 8: Parking Lot Lighting Forecast

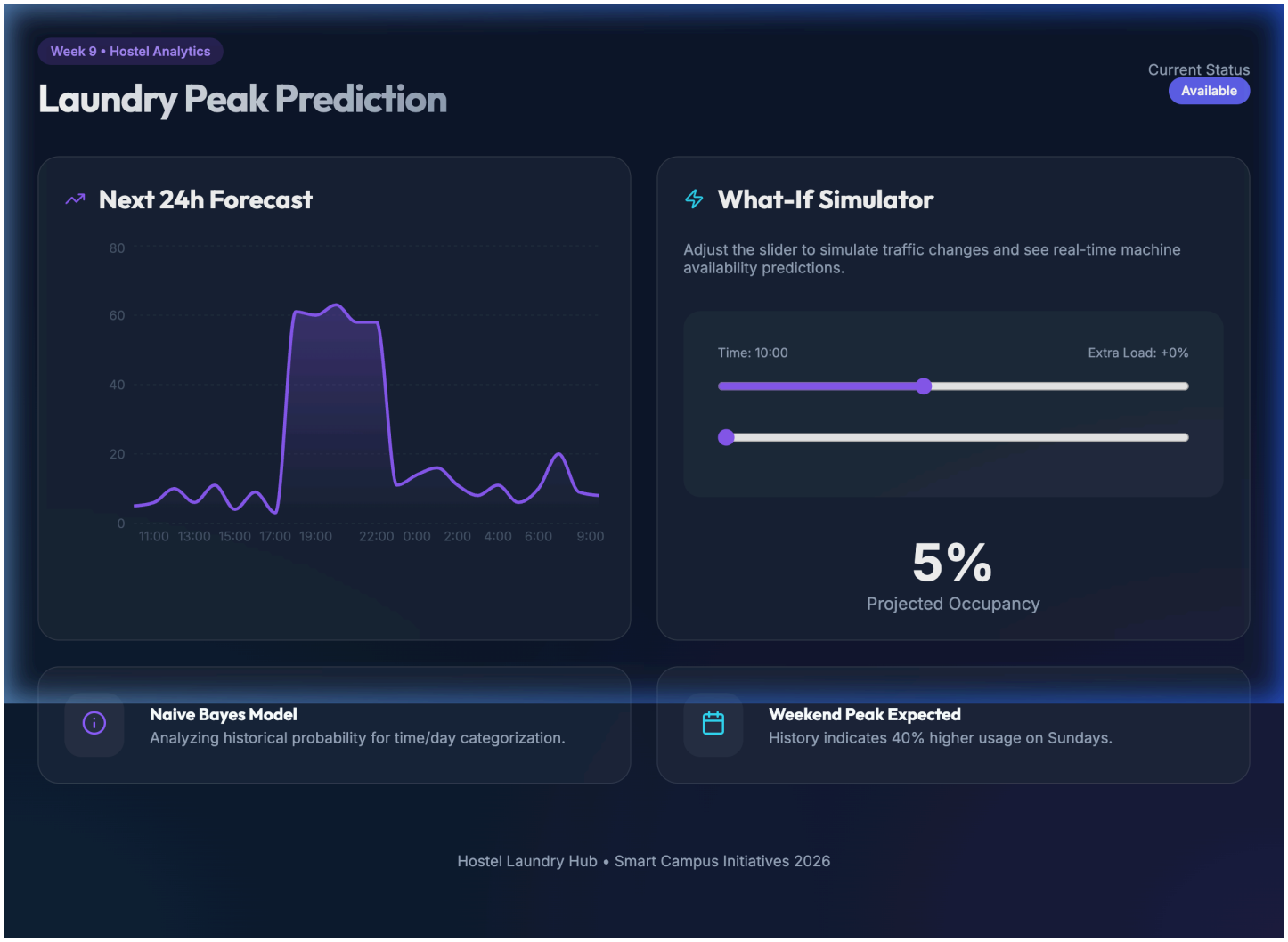
Uses sensor-based vehicle count data with polynomial regression for light usage. Real-time bar chart with anomaly alerts.

Repository: https://github.com/DevDhapodkar/Parking-Lot-Lighting-Forecast

Week 9: Hostel Laundry Peak Prediction

Time-series data with naive Bayes for usage categories; forecasting with Prophet. Dashboard with timeline slider for what-if scenarios.

Repository: https://github.com/DevDhapodkar/Hostel-Laundry-Peak-Prediction

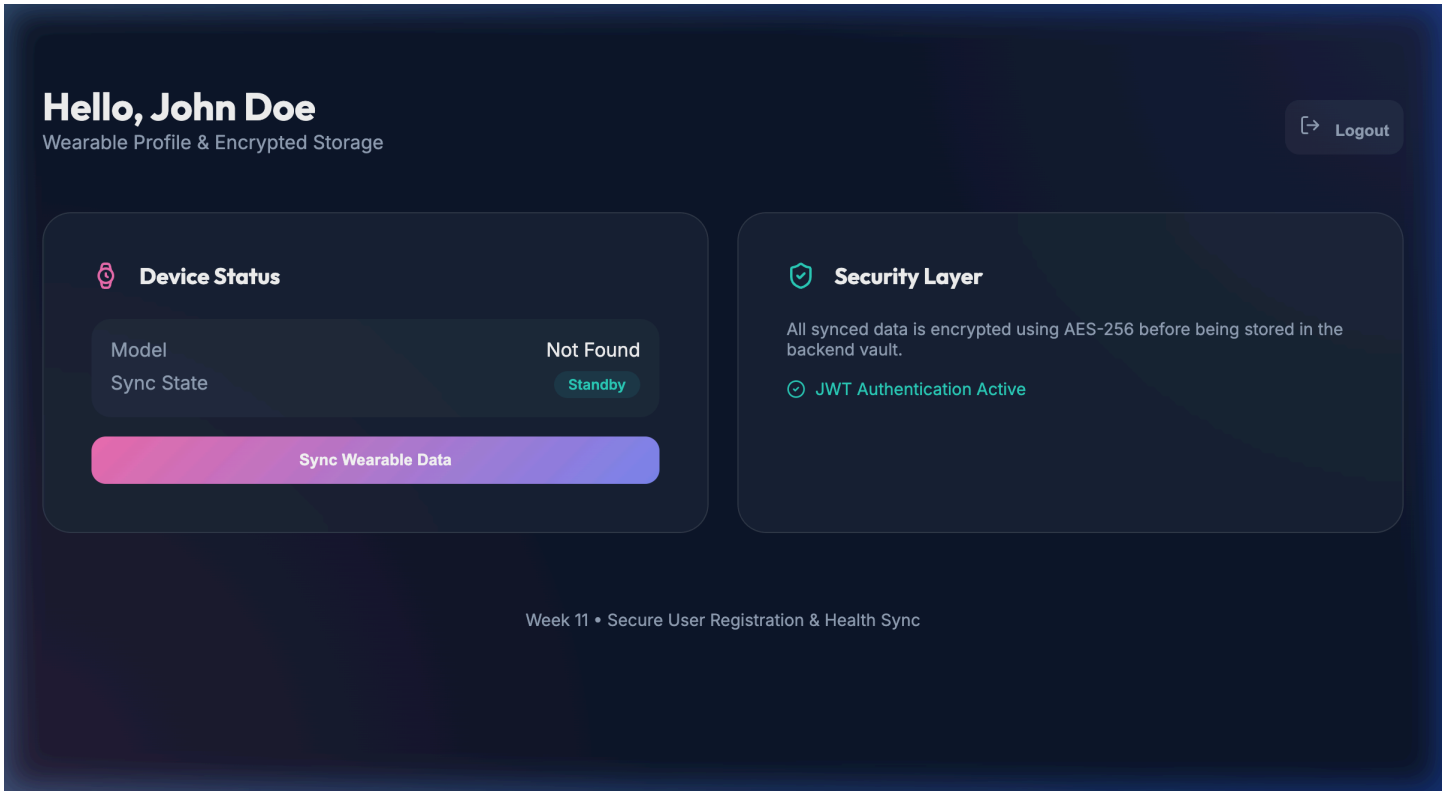


Week 10: Campus-Wide Sustainability Tracker

Ensemble basic models (regression + smoothing) on aggregated data. Comprehensive dashboard with KPIs like carbon savings.

Repository: https://github.com/DevDhapodkar/Campus-Wide-Sustainability-Tracker

Week 11: User Registration Portal

Secure wearable data management with JWT authentication and AES-256 encryption. React interface for synchronization and profile decryption.

Repository: https://github.com/DevDhapodkar/User-Registration-Portal